

Omnichannel Marketing Campaign & Consumer Behaviour Analysis

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Project Overview

This project explores customer purchasing behaviour and marketing campaign performance using a dataset of 2,205 customer records from an e-commerce retailer. The objective was to clean and transform raw customer data, identify demographic spending patterns, evaluate campaign effectiveness and uncover business insights that can support future marketing decisions.

Tools Used

- Google BigQuery (SQL)
- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers
- GitHub

Data Cleaning & Transformation

The original dataset contained several binary columns and categorical variables that were difficult to analyse directly. Using SQL in Google BigQuery, I transformed the dataset into a more analysis friendly format by creating new descriptive variables.

The data cleaning process included:

- Categorising customers into Low, Medium and High-Income groups.
- Grouping education levels into simplified education categories.
- Creating readable marital status labels.
- Creating household categories based on the presence of children and teenagers.
- Converting campaign responses from binary values into Accepted and No Response.
- Preparing a clean analytical dataset for Excel reporting.

```
18 ELSE 'Unknown',
19 END AS Clean_Education_Level,
20 CASE WHEN AcceptedCmp1 = 1 THEN 'Accepted' ELSE 'No Response'
21 END AS Campaign_1,
22 CASE WHEN AcceptedCmp2 = 1 THEN 'Accepted' ELSE 'No Response'
23 END AS Campaign_2,
24 CASE WHEN AcceptedCmp3 = 1 THEN 'Accepted' ELSE 'No Response'
25 END AS Campaign_3,
26 CASE WHEN AcceptedCmp4 = 1 THEN 'Accepted' ELSE 'No Response'
27 END AS Campaign_4,
28 CASE WHEN AcceptedCmp5 = 1 THEN 'Accepted' ELSE 'No Response'
29 END AS Campaign_5,
30 CASE WHEN Kidhome > 0 AND Teenhome = 0 THEN 'Small kids only'
31 WHEN Kidhome = 0 AND Teenhome > 0 THEN 'Teenagers Only'
32 WHEN Kidhome > 0 AND Teenhome > 0 THEN 'Both kids and Teens'
33 ELSE 'No Children'
34 END AS Household_Type,
35 NumWebPurchases,
36 NumCatalogPurchases,
37 NumStorePurchases,
38 MntWines,
39 MntFruits,
40 MntFishProducts,
41 MntMeatProducts,
42 MntSweetProducts,
43 MntGoldProds,
44 FROM `graphite-dynamo-495628-d8.ifood.Marketin_Analysis`
```

This query will process 447.89 KB when run.
Using on-demand processing quota

Google Cloud

My Maps Project

Search (f) for resources, docs, products, and more

Q Search

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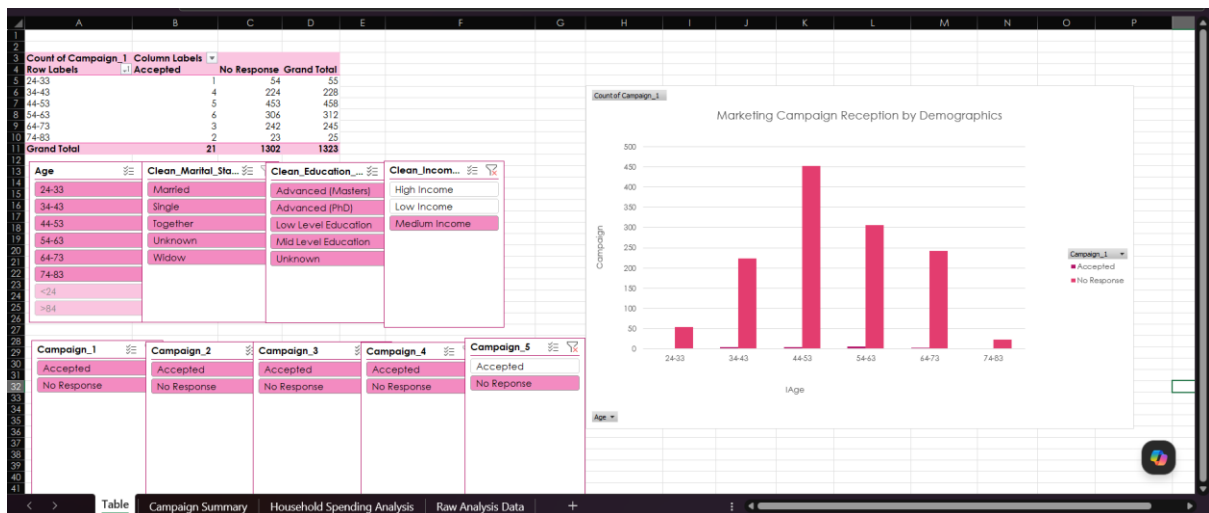
Raw Analytical Dataset

The cleaned dataset served as the foundation for all subsequent analysis. Each customer profile contains demographic information, household classification, purchasing behaviour, and campaign responses, making it easier to analyse customer behaviour across multiple business dimensions.

iFOOD STORE												
Age	Clean_Income_Group	Clean_Marital_Status	Clean_Education_Level	Campaign_1	Campaign_2	Campaign_3	Campaign_4	Campaign_5	Household_Type	NumWebPurchases	NumC	
47	Low Income	Single	Mid Level Education	No Response	No Response	No Response	No Response	No Response	Small kids only	0		
53	Low Income	Unknown	Mid Level Education	No Response	No Response	No Response	No Response	No Response	No Children	0		
49	Low Income	Unknown	Mid Level Education	No Response	No Response	No Response	No Response	No Response	No Children	0		
57	Low Income	Married	Advanced (PHD)	No Response	No Response	No Response	No Response	No Response	Both kids and Teens	0		
44	Low Income	Married	Mid Level Education	No Response	No Response	No Response	No Response	No Response	Teenagers Only	1		
70	Low Income	Together	Advanced (PHD)	No Response	No Response	Accepted	No Response	No Response	Both kids and Teens	1		
52	Medium Income	Together	Mid Level Education	No Response	No Response	No Response	No Response	No Response	Small kids only	2		
36	Low Income	Single	Advanced (Masters)	No Response	No Response	No Response	No Response	No Response	No Children	1		
63	Low Income	Together	Advanced (PHD)	No Response	No Response	No Response	No Response	No Response	Teenagers Only	0		
54	Low Income	Single	Advanced (PHD)	No Response	No Response	No Response	No Response	No Response	Teenagers Only	23		
34	Low Income	Single	Advanced (Masters)	No Response	No Response	No Response	No Response	No Response	Small kids only	5		
29	Low Income	Single	Mid Level Education	No Response	No Response	No Response	No Response	No Response	No Children	0		
49	Medium Income	Together	Advanced (Masters)	No Response	No Response	No Response	No Response	No Response	Small kids only	9		
58	High Income	Married	Mid Level Education	No Response	No Response	No Response	No Response	No Response	No Children	2		
70	High Income	Widow	Advanced (PHD)	No Response	No Response	No Response	No Response	No Response	No Children	1		
68	High Income	Together	Advanced (PHD)	No Response	No Response	No Response	No Response	No Response	No Children	2		
43	High Income	Single	Mid Level Education	No Response	No Response	No Response	No Response	No Response	No Children	3		
60	High Income	Together	Unknown	No Response	No Response	No Response	No Response	No Response	No Children	3		
51	Low Income	Married	Mid Level Education	No Response	No Response	No Response	No Response	No Response	Teenagers Only	25		
69	Medium Income	Unknown	Advanced (Masters)	No Response	No Response	No Response	No Response	No Response	Teenagers Only	1		
60	Medium Income	Married	Advanced (Masters)	No Response	No Response	No Response	No Response	No Response	Teenagers Only	1		
75	High Income	Single	Advanced (PHD)	No Response	No Response	No Response	No Response	No Response	No Children	27		
62	Medium Income	Married	Mid Level Education	No Response	No Response	No Response	No Response	No Response	No Children	2		
48	Medium Income	Married	Advanced (PHD)	No Response	No Response	No Response	No Response	No Response	No Children	3		
55	High Income	Together	Unknown	No Response	No Response	No Response	No Response	No Response	No Children	3		
34	Medium Income	Married	Unknown	No Response	No Response	No Response	No Response	No Response	No Children	2		
41	High Income	Single	Mid Level Education	No Response	No Response	No Response	No Response	No Response	No Children	1		
65	Medium Income	Together	Mid Level Education	No Response	No Response	No Response	No Response	No Response	Teenagers Only	2		
55	High Income	Married	Unknown	No Response	No Response	No Response	No Response	No Response	No Children	4		
65	Medium Income	Single	Unknown	No Response	No Response	No Response	No Response	No Response	No Children	2		

Campaign Performance Dashboard

This dashboard evaluates the performance of five marketing campaigns across different customer segments. The interactive slicers allow for campaign performance to be explored by age, marital status, education level and income group.



Key Insights

- Campaign 1 performed exceptionally well among high-income customers.
- Campaign 3 achieved the most balanced performance across multiple demographic groups, especially amongst low level education, low income and the single.
- Campaign 4 generated moderate engagement across most customer segments.
- Campaign 5 recorded no successful responses, suggesting the campaign strategy requires further investigation.

Campaign Performance by Demographics

The summary tables compare campaign acceptance rates across different demographic characteristics.

Table 1: Campaign Analysis by Marital Status						
Campaign	Single	Married	Together	Widow	Unknown	
Campaign 1	3.37%	4.31%	3.05%	4.35%	0%	
Campaign 2	0.45%	0.63%	0.95%	0%	0.46%	
Campaign 3	8.31%	6.22%	6.29%	4.35%	7.83%	
Campaign 4	4.49%	5.46%	4.57%	7.25%	5.99%	
Campaign 5	0%	0%	0%	0%	0%	

Table 2: Campaign Analysis by Income group			
Campaign	Low Income	Medium Income	High Income
Campaign 1	0%	1.53%	15.10%
Campaign 2	0%	0.63%	0.57%
Campaign 3	7.03%	6.35%	5.38%
Campaign 4	0%	6.42%	5.70%
Campaign 5	0%	0%	0%

Table 3: Campaign Analysis by Education Level					
Campaign	Low Level Education	Medium Level Education	Advanced (Masters)	Advanced(PhD)	Unknown
Campaign 1	0%	3.80%	2.67%	3.20%	6.38%
Campaign 2	0%	0.49%	0.30%	1.20%	0.53%
Campaign 3	11%	6.33%	5.64%	8.22%	6.91%
Campaign 4	0%	4.67%	6.23%	6.62%	3.72%
Campaign 5	0%	0%	0%	0%	0%

Table 4: Campaign Analysis by Age							
Campaign	24-33	34-43 years	44-53 years	54-63 years	64-73 years	74-83 years	
Campaign 1	3.34%	3.14%	1.78%	5.30%	4.65%	12.12%	
Campaign 2	0%	0.44%	1.03%	0.64%	1.22%	0%	
Campaign 3	3.64%	10.53%	7.21%	4.17%	7.35%	8%	
Campaign 4	1.82%	3.51%	5.68%	9.94%	6.94%	8%	
Campaign 5	0%	0%	0%	0%	0%	0%	

Marital Status

Campaign performance remained relatively consistent across marital groups, although Campaign 3 generally produced stronger engagement than the remaining campaigns.

Income Groups

High-income customers showed the strongest response to Campaign 1, while Campaign 3 achieved better engagement among medium- and low-income customers. This suggests that different customer income segments respond differently to marketing strategies.

Education Levels

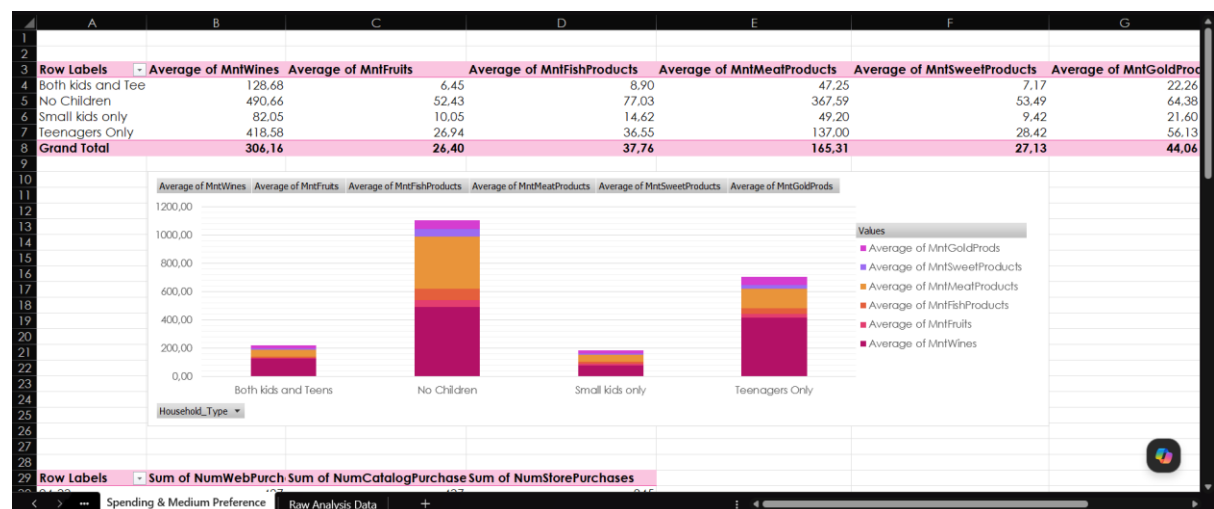
Customers with lower education levels responded particularly well to Campaign 3, while Campaign 1 performed better among customers with higher incomes rather than specific education groups.

Age Groups

Campaign 1 was received quite well by the 74–83-year-olds. Campaign 2 did not perform so well throughout all age groups. Campaign 3 was received relatively well by all age groups with Campaign 4 being a hit for 54-63 year-olds.

Household Spending Analysis

This analysis compares average spending across household types.



Customers with no children consistently recorded the highest spending across most product categories, particularly wine and meat products. Households with small children demonstrated considerably lower spending and spending increases with households with teenagers only, suggesting that family responsibilities may influence purchasing behaviour.

Income Group Spending Analysis

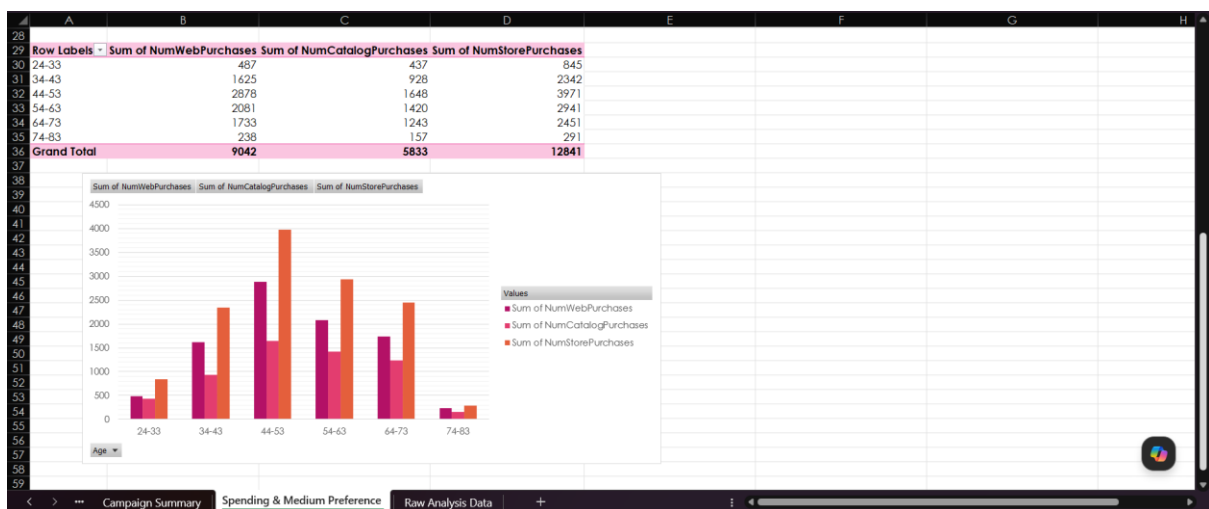
Average spending was analysed across different income groups to identify purchasing patterns.



High-income customers consistently spent more across all products, spending mostly on wine and meat product categories, while low-income customers recorded the lowest average spending. Medium-income customers displayed moderate purchasing behaviour across most products.

Purchase Channel Preference

Customer purchasing activity was analysed across web, catalogue and physical store purchases.



Physical stores remained the dominant purchasing channel across nearly every age group. Web purchases ranked second, while catalogue purchases generated the lowest level of customer activity.

Business Recommendations

I would advise iFood Store of the following:

- Prioritise Campaigns 1 and 3 for future marketing investment.
- Investigate the reasons behind Campaign 5's poor performance before launching similar initiatives.
- Increase marketing efforts towards middle-aged customer segments, which generated the strongest engagement.
- Continue investing in physical retail channels while strengthening digital marketing initiatives to encourage additional online purchases.
- Develop targeted campaigns for different income groups rather than using a single campaign for all customers.

Conclusion

This project demonstrates an end-to-end marketing analytics workflow, from SQL data cleaning and feature engineering to dashboard development and business insight generation. By combining Google BigQuery with Microsoft Excel, the analysis transformed raw customer data into meaningful business intelligence that can support strategic marketing decisions.